



Turtle Tech
Shengji Gene
SAINT GENOMICS

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BioDigital 青

BioDigital-QING Digital PCR

Truly automatic droplet generation digital PCR / Original high-performance glass-based microfluidic chip



Product Highlights

- **Original glass-based microfluidic chip**

Higher thermal conductivity, more uniform reaction, higher light transmittance and better signal quality

- **Solid oil physical isolation technology**

More stable droplets, with an effective droplet ratio of over 95%

- **Truly automatic droplet generation digital PCR**

No manual sampling loading required

- **Extremely low startup cost**

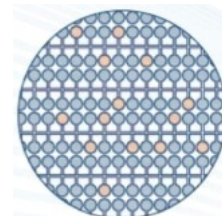
Can start with as few as one sample, eliminating the need to accumulate samples for testing

- **In-situ sample cooling**

Ensure the quality of nucleic acid samples

- **High throughput, high performance**

Can process up to 96 samples at a time, with 4-5 fluorescence channels and 20,000 or more droplets



Application Fields



Tumor liquid biopsy



Pathogen detection



Reproductive health and genetic diseases



Personalized medication guidance



Agriculture and food safety



Nucleic acid detection quality control/verification



Workflow

Fully Automated
Droplet Generation



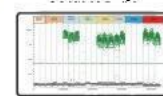
Chip PCR Amplification



Chip Fluorescence Signal Reading



Data Analysis



Automated report generation



BioDigital-QING Digital PCR

✓ Fully Automated Sample Processing System Loader Z200

Initial sample loading volume	25~30μL
Throughput	1~96 samples, fully automated
Number of droplets	> 20,000
Effective droplet ratio	> 95%
Droplet uniformity (CV%)	< 1%
Preparation time	< 2 minutes/sample
QR code scanning function	Available
Interconnection function	Support upgrade



Catalog number: B004
Xiang-Zhu Medical Device Filing No. 20210021

✓ PCR Amplifier Cyclor Z200

Throughput	1~24 samples
4°C incubation	Unlimited duration
Temperature accuracy	≤ ±0.2°C
Amplification time	1.5 hours
Amplification modules	2



Catalog number: B005
Xiang-Zhu Medical Device Registration Approval No. 20212220277

✓ Biochip Reader Imager Z200

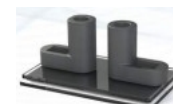
Fluorescence channels	FAM, HEX(VIC), ROX, Cy5, custom channels
Throughput	1~24 samples
Linear dynamic range	5 logs
Minor mutation detection accuracy	0.01%
Reading time	< 2 minutes/sample



Catalog number: B006
Xiang-Zhu Medical Device Registration Approval No. 20212220236

✓ Droplet Generation Chip

Chip 20K-B	20,000 droplets
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✓ Digital PCR Universal Kit

Kit Composition	Details
Reagents	Universal PCR MIX, Oil Phase A & B
Consumables	Droplet generation chip, 8-tube strip/96-well plate/single PCR tube, oil tank, pipette tips



Catalog number: A001
Su-Xu Medical Device Filing No. 20210020

✓ Tumor Personalized Medication Detection Kits

Disease	Kit Name	Specification	Corresponding Targeted Drugs	Catalog No.
Non-small cell lung cancer	Human EGFR gene L858R and 19del mutation detection kit	24T	Gefitinib, erlotinib, icotinib	A007-02
	Human EGFR gene T790M/C797S transversion mutation detection kit	24T	Prediction of drug resistance in combination therapy with first and third generation EGFR-TKI	A011-02
	Human BRAF gene V600E mutation detection kit	24T	Darafenib + trametinib, velotinib	A008-02
	Human KRAS gene mutation detection kit	24T	Prediction of drug resistance to EGFR-TKI	A022-01
	Human PIK3CA gene mutation detection kit	24T	Prediction of resistance to EGFR-TKI and anti-HER2 drugs	A023-01
	Human HER2 gene mutation detection kit	24T	Pozitinib, afatinib, trastuzumab	A017-02
	Human NRAS gene mutation detection kit	24T	Prediction of drug resistance to EGFR-TKI	A016-02
	Human EML4ALK gene fusion mutation detection kit	24T	Ketuzelitinib, serizitinib, aletinib, buccatinib, loratetinib	A013-02
	Human ROS1 gene fusion mutation detection kit	24T	Kizolitinib, seretinib	A014-02
Colorectal cancer	Human NRAS gene mutation detection kit	24T	Prediction of resistance to anti-EGFR antibody drugs	A016-02
	Human KRAS gene mutation detection kit	24T	Prediction of resistance to anti-EGFR antibody drugs	A022-01
	Human PIK3CA gene mutation detection kit	24T	Prediction of resistance to EGFR-TKI and anti-HER2 drugs	A023-01
	Human BRAF gene V600E mutation detection kit	24T	Darafenib + trametinib, velotinib	A008-02
Gastrointestinal stromal tumor	Human c-KIT gene mutation detection kit	24T	Imatinib, Sunitinib	A024-01
	Human PDGFRA gene mutation detection kit	24T	Cetoparib, vicetib, panitamb	A025-01
Hematological tumor	Human JAK2 gene V617F mutation detection kit	24T	Pamidimab, Ruxolitinib	A021-01
	Human c-KIT gene mutation detection kit	24T	Imatinib, Sunitinib	A024-01
	Human BCR-ABL1 gene fusion mutation detection kit	24T	Imatinib, Dasatinib	A019-01
	Human ABL1 gene T3151 mutation detection kit	24T	Imatinib, Ponatinib	A020-01

✓ Pathogenic Microorganism Detection Kit

Kit Name	Specification	sensitivity	Catalog No.
Novel Coronavirus (SARS-CoV-2) Nucleic Acid Detection Kit	24T	100 copies/mL	A005-02
Hepatitis B Virus (HBV) Nucleic Acid Detection Kit	24T	30 copies/mL	A027-01
Hepatitis C Virus (HCV) Ultra-sensitive Nucleic Acid Detection Kit	24T	30 copies/mL	A028-01
Human Immunodeficiency Virus (HIV-1) Ultra-sensitive Nucleic Acid	24T	30 copies/mL	A015-02
Mycobacterium Tuberculosis (TB) Nucleic Acid Detection Kit Human	24T	30 copies/mL	A029-01
Cytomegalovirus (HCMV) Nucleic Acid Detection Kit	24T	30 copies/mL	A031-01
Hepatitis B Virus (HBV), Hepatitis C Virus (HCV) and Human Immunodeficiency Virus (HIV) Nucleic Acid Detection Kit	24T	30 copies/mL	A012-02

✓ Preconception and Prenatal Screening Kit

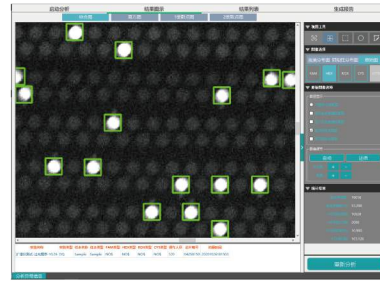
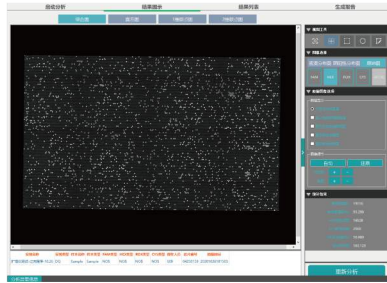
Kit Name	Specification	Corresponding Drug	Catalog No.
Human Survival Motor Neuron 1 (SMN) Exon Deletion Detection Kit	24T	Nusinersen	A033-01
Human Fetal Chromosomal Aneuploidy (T21, T18, T13) Gene Detection Kit	24T	/	Provided by partners



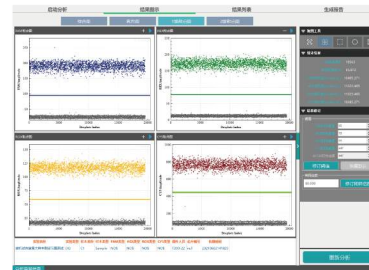
Detection Diagram



Trace original droplets, with quality control accurate to individual droplet



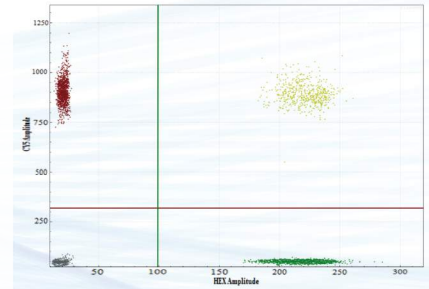
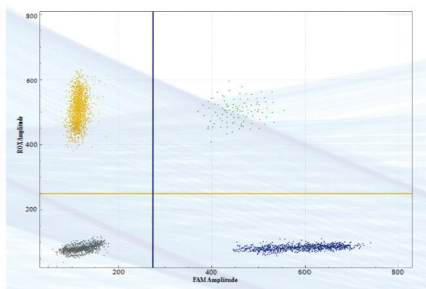
Diversified data presentation



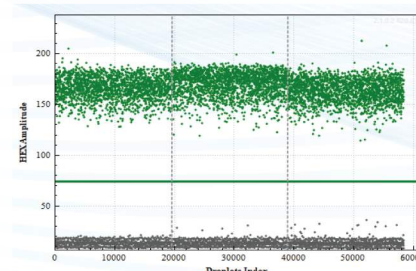
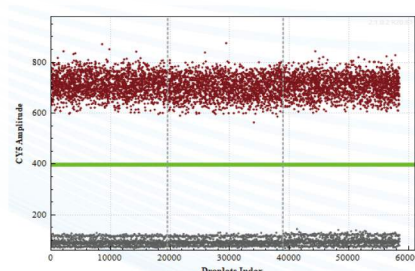
Performance Demonstration



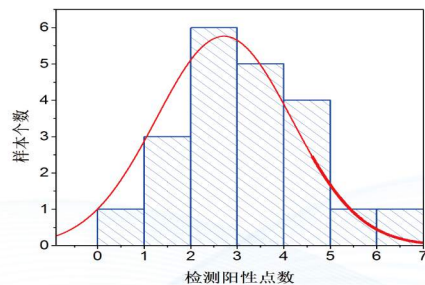
Four fluorescence channels are clearly distinguishable



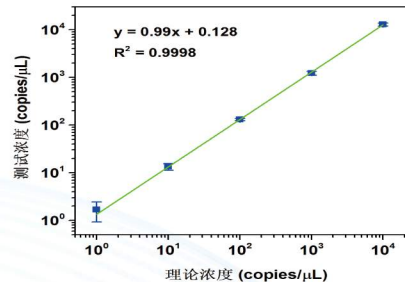
High signal-to-noise ratio and good signal repeatability



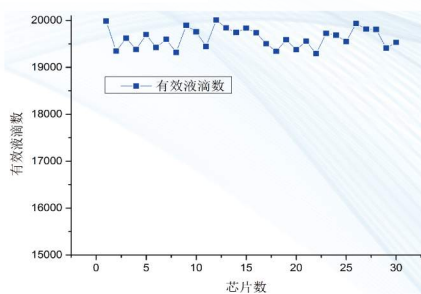
✓ Sensitivity Example



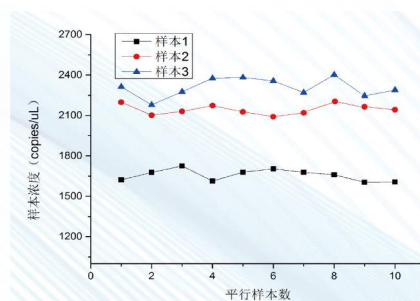
✓ Accuracy and Linear Range Example



✓ Effective Droplet Number Example

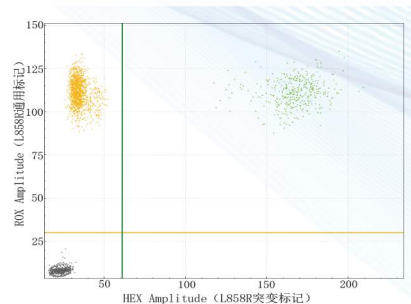
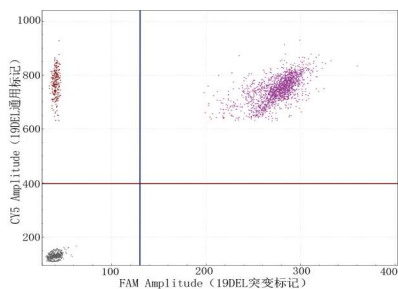


✓ Repeatability and CV Value Example

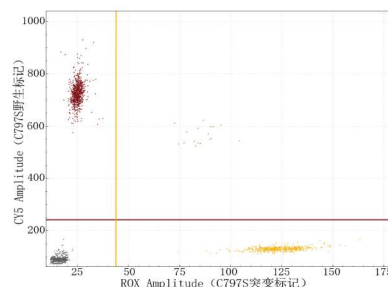
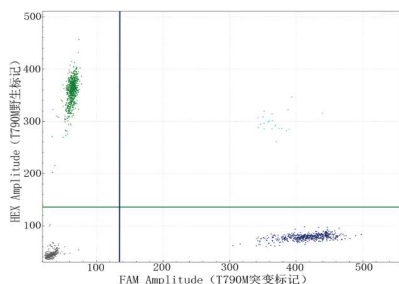


Test Report

1 EGFR 19DEL L858R Detection Example

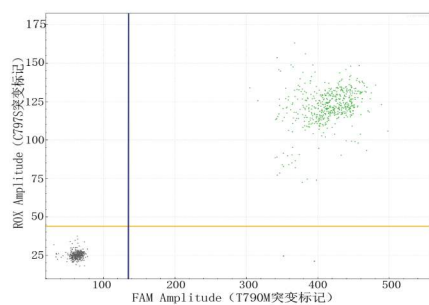


2 T790M C797S Detection Example

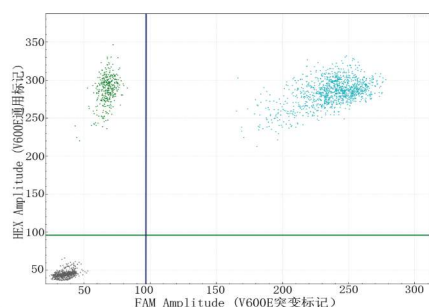




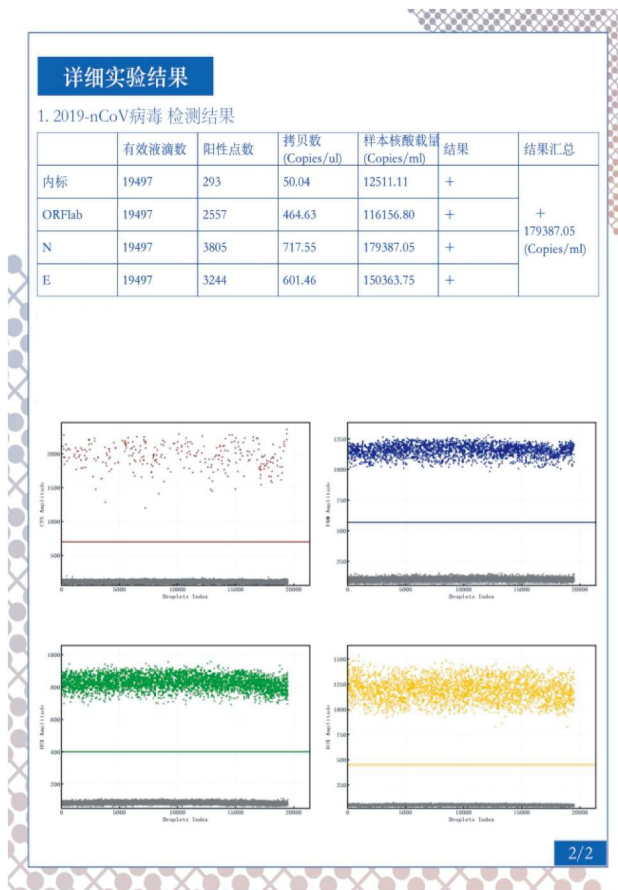
Test Report



3 BRAF V600E Detection Example



4 Novel Coronavirus (2019-nCoV) Detection Example



Credentials

